

Modeling & Simulating the Lake Mburo National Park Trip

Evaluating Entry Gate Queueing Delays
for the 60th Anniversary Surge, By

EZAMAMTI RONALD AUSTINE

Registration No: S23B23-018

Access No: B24252

THE CORE PROBLEM

The Surge

Lake Mburo National Park is preparing for a 30% increase in tourist traffic for its 60th anniversary celebrations.

The Bottleneck

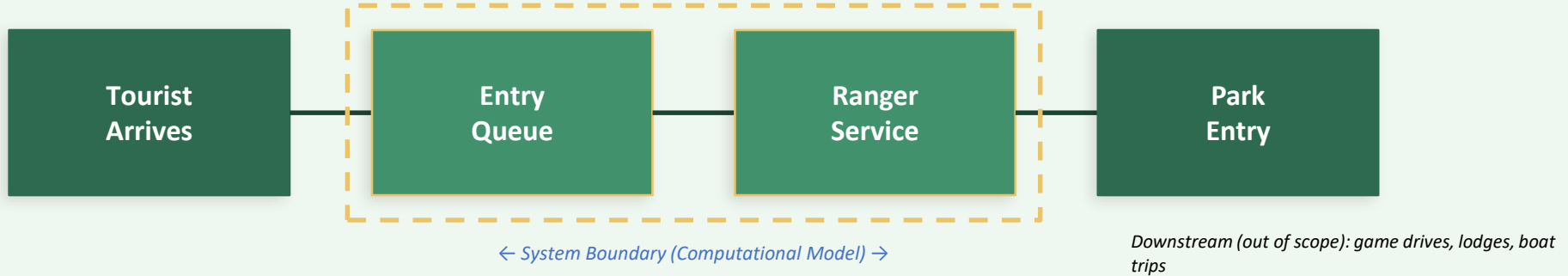
The entry gate is a single-server queue. One ranger processes one vehicle at a time — the primary delay point.

The Objective

Evaluate queueing delay as arrival intensity (λ) grows from 0.50 to 0.65 tourists/min across four service models.

Service capacity: $\mu \approx 0.666$ vehicles/min (mean service time ≈ 1.5 min)

CONCEPTUAL MODEL & ASSUMPTIONS



Key Assumptions:

- Single-server FIFO queue; multi-booth behavior excluded
- Poisson arrivals — $\lambda \in \{0.50, 0.55, 0.60, 0.65\}$ tourists/min
- Each experiment: 10,000 simulated vehicles
- Non-preemptive first-come-first-served discipline
- System starts empty (idle server, empty queue at $t = 0$)

SIMULATION LOGIC: EVENT SCHEDULING

State Variables & Engine

server_busy — binary flag: is the ranger serving a vehicle right now?

waiting_queue — FIFO list storing arrival times of vehicles currently waiting.

Future Event List (FEL) — sorted list governing chronological event processing.

Implementation: Trace-driven engine written in standard C.

Event Handlers

ARRIVAL EVENT:

If server idle → begin service immediately, schedule departure.

If server busy → store arrival time in queue.

DEPARTURE EVENT:

If queue empty → server becomes idle.

If queue has vehicles → immediately serve next; compute queueing delay = clock – stored arrival time.

SERVICE MODELS M/G/1: INVERSE TRANSFORM METHOD

10,000 service values pre-generated per scenario via trace_generator.c for full reproducibility.

01

Deterministic (M/D/1)

Fixed 1.5 min per vehicle — the baseline scenario. Zero variability.

02

Exponential (M/M/1)

Classic baseline. $U \sim (0,1)$ transformed via $-\ln(U) / \mu$. Standard memoryless service.

03

Hyper-Exponential (M/G/1)

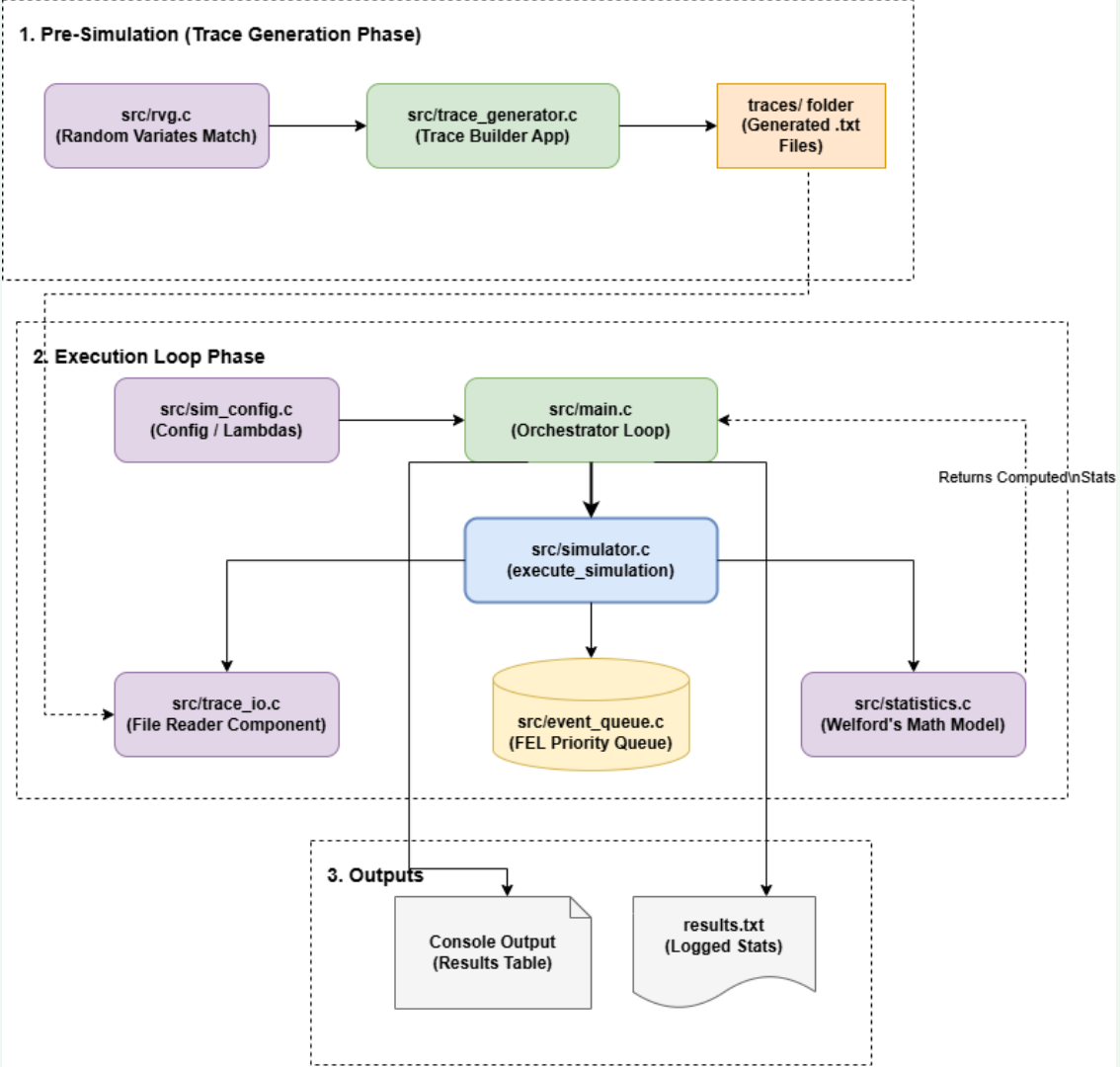
50/50 Bernoulli choice between two exponential means. Models high-variance service.

04

Correlated Exponential (M/G/1)

Bernoulli-copy process introducing positive lag-1 dependence. Simulates 'busy spells'.

System architecture



RESULTS: MEAN QUEUEING DELAY vs λ

λ	Service Model	Mean Delay (min)	Std Dev (min)	Utilization ρ
0.50	Deterministic (1.5)	2.13	2.44	0.7533
0.50	Exponential (M/M/1)	4.32	5.59	0.7565
0.50	Hyper-Exponential	4.57	5.73	0.7542
0.50	Correlated Exponential	5.28	6.73	0.7566
0.55	Deterministic (1.5)	3.70	4.17	0.8274
0.55	Exponential (M/M/1)	7.46	8.91	0.8309
0.55	Hyper-Exponential	6.94	7.78	0.8284
0.55	Correlated Exponential	9.55	11.57	0.8306
0.60	Deterministic (1.5)	7.39	8.83	0.8962
0.60	Exponential (M/M/1)	13.33	14.59	0.9001
0.60	Hyper-Exponential	18.20	21.70	0.8951
0.60	Correlated Exponential	18.10	20.53	0.8982
0.65	Deterministic (1.5)	16.43	13.94	0.9638
0.65	Exponential (M/M/1)	46.64	31.36	0.9672
0.65	Hyper-Exponential	35.76	28.76	0.9613
0.65	Correlated Exponential	39.62	35.50	0.9647

VERIFICATION & VALIDATION

Verification

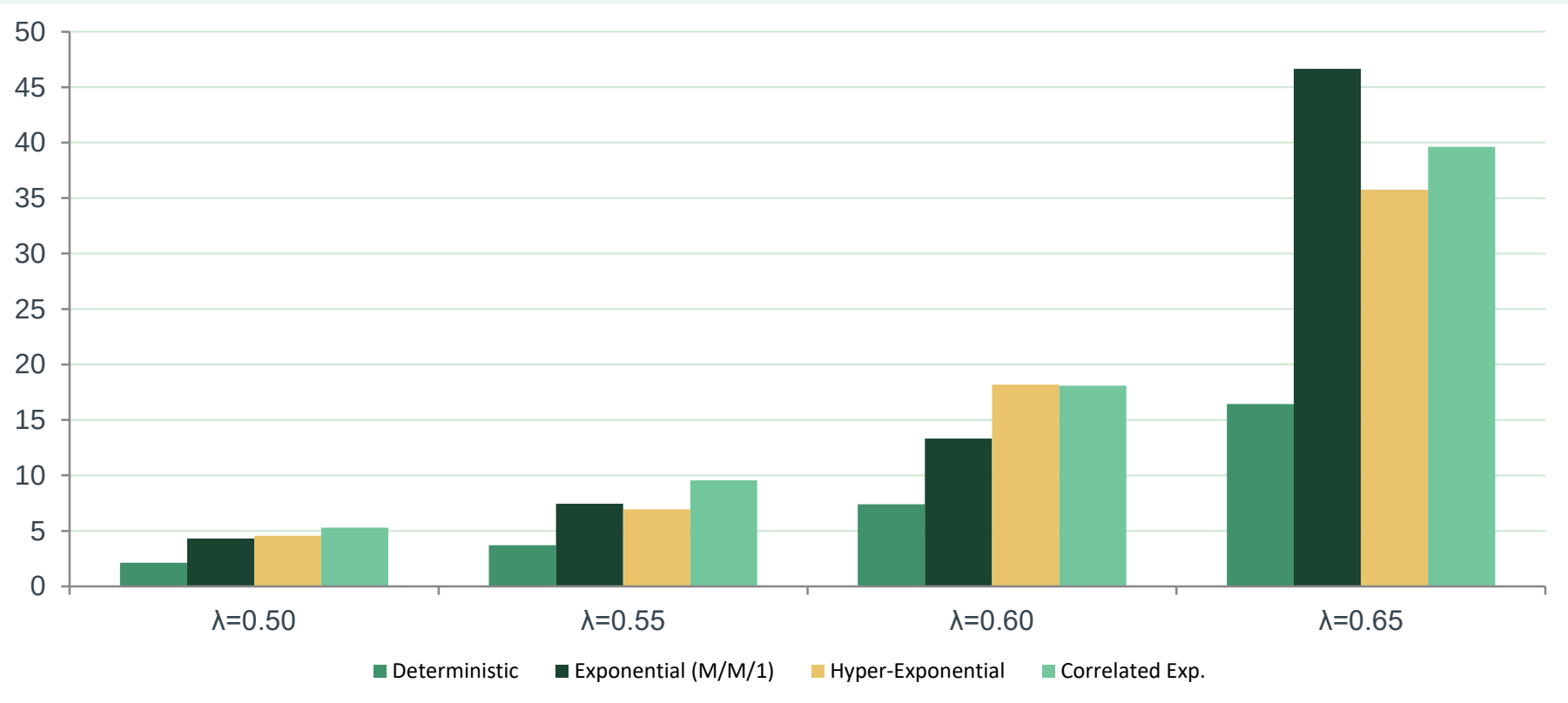
- Debug mode prints first 10 FEL events to console.
- Event ordering and queue transitions are manually inspectable.
- Confirms correct chronological event scheduling.
- Validates FIFO queue state transitions on every departure.

Validation (Theoretical Checks)

- Welford's Algorithm computes running mean & standard deviation.
- Measured utilization compared against $\rho = \lambda / \mu$.
- At $\lambda = 0.50$: measured $\rho = 0.7533$ vs. theoretical ≈ 0.75 . ✓
- All 16 scenarios show consistent utilization near theoretical values.

$$\rho = \lambda / \mu$$

QUEUEING GROWTH: MEAN DELAY vs ARRIVAL RATE



INTERPRETING QUEUEING GROWTH

2.1 min

Det. delay @ $\lambda=0.50$

18.1 min

Hyper-Exp. delay @ $\lambda=0.60$

46.6 min

Exp. delay @ $\lambda=0.65$ ⚠

Key Findings:

- As λ increases from 0.50 $\rightarrow$ 0.65, the arrival rate closes in on service capacity ($\mu \approx 0.666$), causing exponential delay growth.
- At $\lambda = 0.60$: deterministic model yields the lowest delay (7.4 min). Hyper-exponential and correlated-exponential models converge near 18.1 min.
- At $\lambda = 0.65$: the exponential (M/M/1) model spikes to 46.6 min — the highest value recorded — revealing extreme sensitivity near saturation.
- Deterministic service consistently produces the lowest delays, confirming that reducing variability is a key operational lever.

CONCLUSION & RECOMMENDATIONS

⚠ System Threat: The entry gate becomes critically congested as $\lambda \rightarrow 0.65$ tourists/min, with M/M/1 delays reaching 46.6 minutes — far beyond acceptable wait times.

Recommendations:

- Reduce service-time variability — a ranger script or digital processing reduces hyper-exponential scenarios.
- Open a second service lane before the anniversary surge — the M/M/2 configuration should be modeled as a follow-up.
- Implement timed entry slots — pre-booking tourists smooths the Poisson arrival burst.
- Monitor utilization (ρ) in real time — any value approaching 0.90 is a warning threshold.

Thank You

Lake Mburo National Park Simulation | 60th Anniversary Surge Analysis